SUBHAJIT MONDAL | 25EE65R14

SIGNAL PROCESSING AND MACHINE LEARNING

EDUCATION

Year	Degree/Exam	Institute	CGPA/Marks
2027	M.TECH	IIT Kharagpur	8.81 / 10
2023	B.E.	JADAVPUR UNIVERSITY	9.1 / 10
2020	DIPLOMA	WBSCTVESD	93.2%
2015	SECONDARY EXAMINATION	WBBSE	84.2%

SKILLS AND EXPERTISE

- **Languages:** Embedded C, C++, Python, PyTorch
- **AI/ML:** CNN, Vision Transformer, INT8 Quantization
- **Architectures:** ARM Cortex-M
- **RTOS:** FreeRTOS Tasks, Scheduling, Interrupts, Queues
- **Embedded OS:** Bare-Metal Development
- **Protocols:** UART, SPI, I2C, AXI
- **Development Tools:** Vivado, Vitis, Oscilloscope, JTAG Debugging
- **Specializations:** FPGA Co-Design (Zynq), STM32, ESP32, Real-Time Embedded Systems

PROJECTS

Parkinson's Disease Identification using Vision Transformer | FPGA &Edge AI

- Processed five distinct hand-drawing patterns from the **IEEE DataPort dataset** to train a custom, hardware-optimized Vision Transformer (**ViT**) for Parkinson's disease binary classification, achieving **94%** accuracy.
- Designed a custom **ViT accelerator IP** using **Vitis HLS** and applied **INT8** Quantization-Aware Training **(QAT)** optimizing matrix multiplications and AXI memory interfaces to reduce the model footprint by **3x** and maximize throughput.
- Deployed the complete hardware-software co-design on Xilinx Zynq-7000 and ZCU102 SoCs, achieving **1 ms inference latency** and a **60% speedup** compared to standard ARM-only execution.

INTERNSHIPS

Parkinson's Disease Detection via Hand-Drawing Analysis on PYNQ-Z2 FPGA | IIT Kharagpur | May to July 2026

- Trained a MobileNetV2-based transfer learning model on the **HandPD dataset** (736 spiral/meander drawings) for binary PD classification, fine-tuning layers with augmentation (rotation, affine, color jitter), achieving **95.95% accuracy** and **97.39% F1-score**.
- Applied INT8 Quantization-Aware Training (PyTorch FX Graph mode) and exported to **ONNX** (Opset 13) with graph-level operator fusion, compressing model size by **73.3%** (9.15 MB to 2.44 MB) with only **3% accuracy drop** (92.57%).
- Deployed the optimized ONNX model on a Xilinx PYNQ-Z2 (**Zynq-7020**) with a live webcam-based preprocessing pipeline (grayscale, bilateral filtering, CLAHE, Otsu binarization, ROI cropping), achieving real-time diagnosis at **0.73 FPS** and **3 W power consumption**.

COURSEWORK INFORMATION

- **Artificial Intelligence:** ML Techinques, Deep Learning
- **Mathematics:** Linear Algebra, Probability and Random Processes, Convex Optimization
- **Signal Processing:** Statistical Signal Processing, Digital Signal Processing, Signal and System
- **System Design:** Operating System, Computer Architecture

PUBLICATIONS

A message queuing telemetry transport (MQTT) protocol based energy-efficient smart, wireless LED street lighting solution | Journal of Optics | Springer Nature

- Built an **MQTT-based IoT control system** (**Raspberry Pi** publisher, **Mosquitto broker**, **ESP32** subscribers) to wirelessly switch street LEDs across **3 Beam Angle Switching modes** (PRO, COUNTER, FULL-GLOW) based on real-time traffic density, validated across **two hardware phases**.
- Designed **opto-isolated MOSFET** and **Darlington-pair relay driver circuits** to interface microcontroller logic with **12V-55W** LED loads, tested on both a prototype array and a **standard 55W street luminaire**.

POSITIONS OF RESPONSIBILITY

- Assigned as a student volunteer (hardware coordinator) at **Smart India Hackathon (Hardware)**.
- Served as a Teaching Assistant for the **Signal Processing and System Design** Lab, mentoring 40+ faculty trainees during the INAEIIT KGP CEEE program.

EXTRA CURRICULAR ACTIVITIES

- Member, Gym Society, Atal Bihari Vajpayee Hall of Residence, IIT Kharagpur

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